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17th Feb 2026 | CS6370

→ Lexical Semantics

↳ modeling the meanings of words

- words were assumed to be orthogonal to each other (problems with efficiency)

↳ in terms of effectiveness, semantic search is one of the issues

eg. → cat licking mirror

→ animals eating glass.

→ agnostic to meaning.

⇓

but we want to include the meaning in picture.

but what does 'meaning' actual mean?

↳ "purpose"

- tricky issue → circularity issue, we have to somehow depend on lang., but lang. itself is not complete.

PL - Denotational, Operational, Axiomatic semantics

↓ what opn is useful.

⇓ what? (logical way of looking at the things)

- how can we model the meaning of words computationally?

→ meaning of a word w/o any proper context has no meaning actually! (circle of context)

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→ good vs bad, they are somewhere related to each other
- Linguistics -
→ Referential (apple)
→ mental model (justice)
→ lang. as used (words don't have meanings on their own, we have to look at the context & then decide eventually).
- not everything that is grammatically correct, might not have meaning.
it depends on context, bigger content of bigger mental model.
but this is based on rules.
- Chomsky → syntax, grammer, rules
- Roger Schank → memory, how memory deals with lang.
↳ as humans think, like we start thinking just by reading the deadly attack.
(This is also known as the information abstraction.)
- in reality, we don't have fragments. but we put boundaries ⇒ reductionist model
- there should a complement to understand a particular entity. Lang. is effective if it helps to create boundaries.